

```

(kali㉿kali)-[~]
$ mkdir -p ~/CIP-B103-Lab5/{evidence,working,exported,reports,screenshots,scripts}

(kali㉿kali)-[~]
$ cd ~/CIP-B103-Lab5

(kali㉿kali)-[~/CIP-B103-Lab5]
$ pwd
/home/kali/CIP-B103-Lab5

(kali㉿kali)-[~/CIP-B103-Lab5]
$ find . -maxdepth 1 -type d -print
.
./working
./screenshots
./evidence
./scripts
./reports
./exported

```

```

(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo apt install -y wireshark tshark python3-scapy net-tools
[sudo] password for kali:
wireshark is already the newest version (4.6.6-1).
tshark is already the newest version (4.6.6-1).
python3-scapy is already the newest version (2.7.0+dfsg1-1).
net-tools is already the newest version (2.10-2).
The following packages were automatically installed and are no longer required:
  libraw23t64 libvidstab1.1 openjdk-21-jre openjdk-21-jre-headless
Use 'sudo apt autoremove' to remove them.

Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1742

(kali㉿kali)-[~/CIP-B103-Lab5]
$ wget -O evidence/arp.pcap \
'https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap'
--2026-08-01 10:26:01-- https://raw.githubusercontent.com/frankwxu/digital-forensics-lab/main/Networking_Forensics/lab_files/ARP_spoofing/arp.pcap
Resolving raw.githubusercontent.com (raw.githubusercontent.com)... 185.199.109.133, 185.199.110.133, 185.199.111.133, ...
Connecting to raw.githubusercontent.com (raw.githubusercontent.com)|185.199.109.133|:443 ... connect

```

```

HTTP request sent, awaiting response... 200 OK
Length: 1072 (1.0K) [application/octet-stream]
Saving to: 'evidence/arp.pcap'

evidence/arp.pcap  100%[=====>] 1.05K  --.-KB/s  in 0s

2026-08-01 10:26:01 (15.3 MB/s) - 'evidence/arp.pcap' saved [1072/1072]

```

```

(kali㉿kali)-[~/CIP-B103-Lab5]
$ cp --preserve=timestamps evidence/arp.pcap working/arp_working.pcap

(kali㉿kali)-[~/CIP-B103-Lab5]
$ sha256sum evidence/arp.pcap working/arp_working.pcap \
| tee reports/arp_capture_hashes.txt
342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e  evidence/arp.pcap
342a75dc002d090cc7fd108994b6c0c9c8eaa3962cf642159b4507d5615adc3e  working/arp_working.pcap

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip -br address | tee reports/interfaces.txt
lo                UNKNOWN    127.0.0.1/8  ::1/128
eth0              UP        192.168.44.128/24  fe80::fcdb:733e:9de1:56cb/64
docker0          DOWN      172.17.0.1/16

```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip route | tee reports/routes.txt
default via 192.168.44.2 dev eth0 proto dhcp src 192.168.44.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.44.0/24 dev eth0 proto kernel scope link src 192.168.44.128 metric 100

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip neigh show | tee reports/arp_table_initial.txt
192.168.44.254 dev eth0 lladdr 00:50:56:e1:c4:05 STALE
192.168.44.2 dev eth0 lladdr 00:50:56:f4:fd:61 REACHABLE
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ tree
.
├── evidence
│   └── arp.pcap
├── exported
├── reports
│   ├── arp_capture_hashes.txt
│   ├── arp_table_initial.txt
│   ├── interfaces.txt
│   └── routes.txt
├── screenshots
├── scripts
└── working
    └── arp_working.pcap

7 directories, 6 files

(kali㉿kali)-[~/CIP-B103-Lab5]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip -br address
lo UNKNOWN 127.0.0.1/8 ::1/128
eth0 UP 192.168.44.128/24 fe80::fcd8:733e:9de1:56cb/64
docker0 DOWN 172.17.0.1/16

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip route
default via 192.168.44.2 dev eth0 proto dhcp src 192.168.44.128 metric 100
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1 linkdown
192.168.44.0/24 dev eth0 proto kernel scope link src 192.168.44.128 metric 100

(kali㉿kali)-[~/CIP-B103-Lab5]
$ IFACE=eth0
GATEWAY_IP=$(ip route | awk '/default/ {print $3; exit}')

(kali㉿kali)-[~/CIP-B103-Lab5]
$ echo $IFACE
echo $GATEWAY_IP
eth0
192.168.44.2
```

```

(kali㉿kali)-[~/CIP-B103-Lab5]
$ Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'

[4] done sudo tshark -i "$IFACE" -f 'arp' -a duration:20 -w /tmp/normal_arp.pcapng
(kali㉿kali)-[~/CIP-B103-Lab5]
$ ping -c 1 "$GATEWAY_IP"
PING 192.168.44.2 (192.168.44.2) 56(84) bytes of data.
64 bytes from 192.168.44.2: icmp_seq=1 ttl=128 time=0.424 ms

— 192.168.44.2 ping statistics —
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.424/0.424/0.424/0.000 ms

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ls -lh /tmp/normal_arp.pcapng
-rw-r--r-- 1 root root 336 Aug  1 10:38 /tmp/normal_arp.pcapng

(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo ip neigh flush "$GATEWAY_IP" dev "$IFACE"

```

```

(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo tshark -i "$IFACE" -f 'arp' -a duration:20 -w /tmp/normal_arp.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
2

(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo tshark -i "$IFACE" -f 'arp' -a duration:20 -w /tmp/normal_arp.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
12

(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo cp /tmp/normal_arp.pcapng evidence/
sudo chown kali:kali evidence/normal_arp.pcapng

(kali㉿kali)-[~/CIP-B103-Lab5]
$ tshark -r evidence/normal_arp.pcapng -Y arp
1 0.0000000000 VMware_87:06:29 → VMware_f4:fd:61 ARP 42

```

```

Who has 192.168.44.2? Tell 192.168.44.128
2 0.000867778 VMware_f4:fd:61 → VMware_87:06:29 ARP 60
192.168.44.2 is at 00:50:56:f4:fd:61
3 6.796966003 VMware_c0:00:08 → Broadcast ARP 60 Who
has 192.168.44.2? Tell 192.168.44.1
4 7.706171743 VMware_c0:00:08 → Broadcast ARP 60 Who
has 192.168.44.2? Tell 192.168.44.1
5 8.583826839 VMware_c0:00:08 → Broadcast ARP 60 Who
has 192.168.44.2? Tell 192.168.44.1
6 9.486958782 VMware_c0:00:08 → Broadcast ARP 60 Who
has 192.168.44.2? Tell 192.168.44.1
7 10.419525199 VMware_c0:00:08 → Broadcast ARP 60 Wh
o has 192.168.44.2? Tell 192.168.44.1
8 11.289932214 VMware_c0:00:08 → Broadcast ARP 60 Wh
o has 192.168.44.2? Tell 192.168.44.1
9 12.195445674 VMware_c0:00:08 → Broadcast ARP 60 Wh
o has 192.168.44.2? Tell 192.168.44.1
10 13.128577471 VMware_c0:00:08 → Broadcast ARP 60 Wh
o has 192.168.44.2? Tell 192.168.44.1
11 13.991450478 VMware_c0:00:08 → Broadcast ARP 60 Wh
o has 192.168.44.2? Tell 192.168.44.1

```

```

(kali㉿kali)-[~/CIP-B103-Lab5]
$ tshark -r evidence/normal_arp.pcapng -Y 'arp' -T fields \
-e frame.number \
-e frame.time \
-e eth.src \
-e eth.dst \
-e arp.opcode \
-e arp.src.proto_ipv4 \
-e arp.src.hw_mac \
-e arp.dst.proto_ipv4 \
-e arp.dst.hw_mac \
| tee reports/normal_arp_fields.tsv
1      2026-08-01T10:41:26.118105322-0400      00:0c:29:87:06:29      00:50:56:f4:fd:61      1 1
92.168.44.128      00:0c:29:87:06:29      192.168.44.2      00:00:00:00:00:00
2      2026-08-01T10:41:26.118973100-0400      00:50:56:f4:fd:61      00:0c:29:87:06:29      2 1
92.168.44.2      00:50:56:f4:fd:61      192.168.44.128      00:0c:29:87:06:29
3      2026-08-01T10:41:32.915071325-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
4      2026-08-01T10:41:33.824277065-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
5      2026-08-01T10:41:34.701932161-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
6      2026-08-01T10:41:35.605064104-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
7      2026-08-01T10:41:36.537630521-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
8      2026-08-01T10:41:37.408037536-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
9      2026-08-01T10:41:38.313550996-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
10     2026-08-01T10:41:39.246682793-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
11     2026-08-01T10:41:40.109555800-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
12     2026-08-01T10:41:41.010313379-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00

```



```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ cat reports/normal_arp_fields.tsv
1      2026-08-01T10:41:26.118105322-0400      00:0c:29:87:06:29      00:50:56:f4:fd:61      1 1
92.168.44.128      00:0c:29:87:06:29      192.168.44.2      00:00:00:00:00:00
2      2026-08-01T10:41:26.118973100-0400      00:50:56:f4:fd:61      00:0c:29:87:06:29      2 1
92.168.44.2      00:50:56:f4:fd:61      192.168.44.128      00:0c:29:87:06:29
3      2026-08-01T10:41:32.915071325-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
4      2026-08-01T10:41:33.824277065-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
5      2026-08-01T10:41:34.701932161-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
6      2026-08-01T10:41:35.605064104-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
7      2026-08-01T10:41:36.537630521-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
8      2026-08-01T10:41:37.408037536-0400      00:50:56:c0:00:08      ff:ff:ff:ff:ff:ff      1 1
92.168.44.1      00:50:56:c0:00:08      192.168.44.2      00:00:00:00:00:00
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ PCAP=working/arp_working.pcap

(kali㉿kali)-[~/CIP-B103-Lab5]
$ echo $PCAP
working/arp_working.pcap

(kali㉿kali)-[~/CIP-B103-Lab5]
$ tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
  -e frame.number \
  -e frame.time_epoch \
  -e eth.src \
  -e eth.dst \
  -e arp.src.proto_ipv4 \
  -e arp.src.hw_mac \
  -e arp.dst.proto_ipv4 \
  -e arp.dst.hw_mac \
  | tee reports/arp_replies.tsv
4      1678591364.904123642      00:50:56:86:02:65      00:50:56:86:cb:fc      136.160.215.194 00:
50:56:86:02:65      136.160.215.15      00:50:56:86:cb:fc
6      1678591369.943043799      00:50:56:86:cb:fc      00:50:56:86:02:65      136.160.215.15 00:
6      1678591369.943043799      00:50:56:86:cb:fc      00:50:56:86:02:65      136.160.215.15 00:
50:56:86:cb:fc      136.160.215.194      00:50:56:86:02:65
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ tshark -r "$PCAP" -Y 'arp.opcode==2' -T fields \
  -e arp.src.proto_ipv4 \
  -e arp.src.hw_mac \
  | sort | uniq -c | sort -nr \
  | tee reports/ip_mac_claims.txt
1 136.160.215.194 00:50:56:86:02:65
1 136.160.215.15 00:50:56:86:cb:fc

(kali㉿kali)-[~/CIP-B103-Lab5]
$ tshark -r "$PCAP" \
  -Y 'arp.opcode==2 && eth.dst!=ff:ff:ff:ff:ff:ff' \
  -T fields \
  -e frame.number \
  -e frame.time \
  -e arp.src.proto_ipv4 \
  -e arp.src.hw_mac \
  -e eth.dst \
  | tee reports/unicast_arp_replies.tsv
```

```
4      2023-03-11T22:22:44.904123642-0500      136.160.215.194 00:50:56:86:02:65      00:50:56:86
:cb:fc
6      2023-03-11T22:22:49.943043799-0500      136.160.215.15  00:50:56:86:cb:fc      00:50:56:86
:02:65
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ cat reports/ip_mac_claims.txt
1 136.160.215.194 00:50:56:86:02:65
1 136.160.215.15  00:50:56:86:cb:fc
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip -br address
lo                UNKNOWN          127.0.0.1/8 ::1/128
eth0              UP                192.168.44.128/24 fe80::fcd8:733
docker0           DOWN             172.17.0.1/16
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip route
default via 192.168.44.2 dev eth0 proto dhcp src 192.168.44.128
172.17.0.0/16 dev docker0 proto kernel scope link src 172.17.0.1
192.168.44.0/24 dev eth0 proto kernel scope link src 192.168.44.
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip neigh show | tee reports/arp_table_before.txt
192.168.44.254 dev eth0 lladdr 00:50:56:e1:c4:05 STALE
192.168.44.2 dev eth0 lladdr 00:50:56:f4:fd:61 REACHABLE
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ IFACE=eth0
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo tshark -i "$IFACE" -f 'arp' \
-a duration:40 \
-w evidence/controlled_arp_poison.pcapng &
[4] 32346
```

```
(kali㉿kali)-[~/CIP-B103-Lab5]
$ Running as user "root" and group "root". This could be dange
Capturing on 'eth0'
tshark: The file to which the capture would be saved ("evidence/
```

```

(kali㉿kali)-[~/CIP-B103-Lab5]
$ Running as user "root" and group "root". This could be dange
(kali㉿kali)-[~/CIP-B103-Lab5]
6 $

[4] done          sudo tshark -i "$IFACE" -f 'arp' -a duration:4
0 -w
(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo tshark -i "$IFACE" -f 'arp' \
-a duration:40 \
-w /tmp/controlled_arp_poison.pcapng &
[4] 33281

(kali㉿kali)-[~/CIP-B103-Lab5]
$ Running as user "root" and group "root". This could be dange
rous.
Capturing on 'eth0'

[4] done          sudo tshark -i "$IFACE" -f 'arp' -a duration:4
0 -w
(kali㉿kali)-[~/CIP-B103-Lab5]
$ █

```

```

(kali㉿kali)-[~/CIP-B103-Lab5]
$ sudo ip neigh flush all

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip neigh show
192.168.44.2 dev eth0 lladdr 00:50:56:f4:fd:61 REACHABLE

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ping -c 1 192.168.44.2
PING 192.168.44.2 (192.168.44.2) 56(84) bytes of data.
64 bytes from 192.168.44.2: icmp_seq=1 ttl=128 time=0.264 ms

— 192.168.44.2 ping statistics —
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 0.264/0.264/0.264/0.000 ms

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ip neigh show
192.168.44.2 dev eth0 lladdr 00:50:56:f4:fd:61 REACHABLE

(kali㉿kali)-[~/CIP-B103-Lab5]
$ ps aux | grep -E '[a]rp.py|[s]copy' | tee reports/process_check.txt

(kali㉿kali)-[~/CIP-B103-Lab5]

```